

Aarnav Noble

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EDUCATION

University of Waterloo

Waterloo, Ontario

B.A.Sc. in Honours Computer Engineering

Sep 2023 – Apr 2028 (expected)

- Relevant Coursework: Operating Systems, Computer Architecture, Compilers, Databases, Computer Networks, Data Structures & Algorithms, Concurrency, Probability & Statistics, Linear Algebra

EXPERIENCE

Member of Technical Staff Intern

Jan 2026 – Apr 2026

AethexAI

San Francisco, California

- Built the platform's public API and Python SDK from scratch, with **30+** endpoints for mid-call session control
- Managed the production AWS/EKS platform on Kubernetes, Terraform, and CI/CD, scaling to **500+** pods
- Fronted the API with an L7 balancer and UDP media with an L4, targeting pod IPs to drop a routing hop
- Hardened the **sub-500ms** pipeline to **75+** concurrent calls, fixing state races and null crashes under load
- Designed a **5-phase** interrupt state machine spanning **3** transports, ending dropped speech and state corruption
- Rebuilt agent-document RAG across ingestion, concurrency, cross-turn dedup, and grounding over **10K+** chunks
- Built a WebRTC call simulator that runs **200+** scripted call scenarios in CI, interrupts included

AI/ML Developer

May 2025 – Aug 2025

Environment and Climate Change Canada

Burlington, Ontario

- Designed and built a PyTorch GNN, **4** message-passing layers over a **10K+** node graph
- Beat gradient-boosted baselines by **85%**, replacing hand-built spatial features with learned graph structure
- Extended usable output **5x** by generating forecasts at graph nodes with no observed data
- Engineered **50+** spatial-temporal features and cut a full run from hours to minutes by vectorizing and parallelizing
- Ran region-holdout cross-validation and a **100+** run hyperparameter sweep to kill spatial overfitting
- Led ML on a 6-person research team from architecture to eval, presenting results toward a publication

Software Engineering Intern

Sep 2024 – Dec 2024

Toronto Transit Commission (TTC)

Toronto, Ontario

- Built a config-as-code deployment pipeline enforcing one software state across a **10K+** device fleet
- Automated zero-touch bare-metal provisioning with network-boot and Bash, cutting setup roughly **50%**
- Wrote Bash orchestration for staged patch and driver rollouts across **2K+** machines, with auto-retry on failure

Software Developer

Jan 2024 – Apr 2024

Ontario Chamber of Commerce

Toronto, Ontario

- Built **10+** reusable React components with schema-driven config panels, editable by non-devs in a visual editor
- Fed external data into **15+** Shopify Metaobject types from a Kubernetes service, bound to live storefront content
- Secured the app with session-token auth, scoped OAuth, and HMAC-verified webhooks against forged events

PROJECTS

Vestige | *Python, Go, TypeScript, Next.js, OpenTelemetry, ClickHouse* |

- Built a Go service ingesting OpenTelemetry spans in **sub-20ms**, with zero loss through a ClickHouse outage
- Tuned the ClickHouse trace schema for one-read trace loads, and gated CI on a **100-pair** tenant-isolation test
- Built a replay engine that reruns past failures from SHA-256-keyed fixtures in **<90s**, blocking output-changed PRs

Roam | *Python, FastAPI, LightGBM, FAISS, OR-Tools, React Native* |

- Built a React Native app turning **12+** preference signals into a route-optimized, context-grounded itinerary
- Chained FAISS retrieval, LightGBM LambdaRank, and OR-Tools routing into a **4-stage** pre-LLM pipeline
- Trained a LambdaRank ranker on **8** features, benchmarked **24%** over a semantic-similarity baseline

SKILLS

Languages: Python, C/C++, Go, JavaScript, TypeScript, Java, SQL, Bash

Systems & Backend: Kubernetes, Docker, Terraform, AWS, CI/CD, Linux, GraphQL, FastAPI, Node.js, PostgreSQL

ML & Data: PyTorch, NumPy, pandas, Scikit-learn, XGBoost, LightGBM, FAISS, OR-Tools

Concepts: Distributed Systems, System Design, Scalability, Load Balancing, Autoscaling, GPU Orchestration, Model Serving, LLM Inference, RAG, Data Pipelines & ETL, Infrastructure as Code, Observability, API & SDK Design